Lecture 2 - 29/10

Thursday, 29 October 2026 13:19

News

last time we did reductions to SETH, APSP, 3sum and that last 2 are broken.
2days ago SETH is broken

our reduction was SETH → OV → things

doesn't apply ov yet, so for now they stand.

3sum

Given a set S of n integers in $[-U, +U]$. are there $a,b,c \in S$ s.t. $a+b+c=0$?

Algs:
- naive: $O(n^3)$, try every triplet
- $O(n^2 \log n)$ - sort, go over pairs, binary search last one
- $O(n^2)$ rand hash sum of all pairs
- $O(n^2)$ sort, have 2 indexs, one at start one at end, move if its small or big
- $O(n^{1.99})$

K-sum: $a_1 + a_2 + \dots + a_k = 0$

hash sums of $k/2$, select $k/2$, look for its -

$O(n^{\lceil k/2 \rceil})$

$2 \to n$
$3 \to n^2$
$4 \to n^2$
$5 \to n^3$
$6 \to n^3$

this is not broken yet

Variants

3-sum finding: given S... return $a,b,c \in S$ s.t. $a+b+c=0$ if it exists

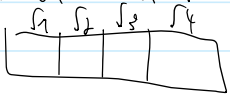
Thm: ("finding → decision"):

if 3-sum is in time $T(n)$ then 3-sum finding is in $O(T(n))$ time

- split randomly, call 3 sum, keep doing until 1 are in your split, $(\frac{1}{2})^3 = \frac{1}{8}$ its in the smaller split

  $O(1) \cdot T_{dec}(n) + T_{find}(n)$ this works but small

- Partition to 4 sets, check every set of 3 of them

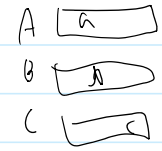


  let $S_i' = S \setminus S_i$
  note: $|S_i'| = \frac{3}{4} n$
  - call 3-sum(S_i') for each i
  - recurse on first find else, return false.
  - if $|S| = O(1)$, solve in $|S|^3$ time

correctness: ✓
time: $T_{find}(n) \le 4\, T_{dec}(n) + T_{find}(\frac{3}{4}n) + O(n) = O\left(n + \sum_{i=0}^{\log n} \left(T_{dec}(\frac{3}{4})^i \cdot n\right)\right) =$

$\le O\left(n + \left(\sum_{i=0}^{\log n} (\frac{3}{4})^i\right) \cdot T_{dec}(n)\right) = O(T_{dec}(n))$

colorful-3sum: Given 3 sets of n numbers, $A, B, C \in [-U, U]$ are there $a \in A, b \in B, c \in C$ s.t. $a+b+c=0$



Thm 2: if col-3sum is in $T(n)$ then there is an alg that solves 3-sum in time $O(T(n))$ and succeeds w.p. $9/10$

IF: - repeat 10 times independently w.p. $\frac{1}{3}$
  - $\forall x \in S$ let $col(x) = \{1, 2, 3\}$
  - let $A = \{x \in S \mid col(x) = 1\}$
    $B = \{\ldots 2\}$
    $C = \{\ldots 3\}$
  - if col-3sum(A,B,C) = "yes" return yes

- return "no"

correctness:

fix a sol $a,b,c \in S$

$\Pr[\text{all 3 sets distinct colors}] = 3!/3^3 = 2/9$

"no" → always "no"

"yes" → "yes" w.p. $> 1/10$

$\Pr[\text{all 10 calls fail}] \leq (7/9)^{10} < \frac{1}{10}$

col-3sum → 3sum

Thm: if 3sum is in $T(n)$ then col-3sum is in $O(T(n))$

A ▭ $+60U$

B ▭ $-40U$

C ▭ $-20U$

$S = \{a+60U \mid a \in A\} \cup \{b-40U \mid b \in B\} \cup \{c-20U \mid c \in C\}$

"yes" → "yes"

"no" → "no" sum of any 3 with repetitions in $\{60, 40, -20\}$ is ≥ 10

3sum': given set S, are there $a,b,c \in S$ s.t. $a+b=c$?

3sum $\xrightarrow{+60U}$ col coding

3sum' $\xrightarrow{+c}$ col coding

col-3sum $\leftrightarrow$ col-3sum'

this is easy, just do $C' = \{-c \mid c \in C\}$

3AVARAGE: given set S, are there $a,b,c \in S$ s.t. $\frac{a+b}{2} = c$?

do same thing, go to colored

AVG $\xrightarrow{\text{col coding}}$ col-AVG

3sum $\xrightarrow{+60U}$ col coding

3sum' $\xrightarrow{+c}$ col coding

col-3sum $\leftrightarrow$ col-3sum'

here the same trick wont work, any elements from the +60U cancel out

fact from additive combinatorics:
there is a subset $M \subseteq [U]$ that is "average free" i.e. no $a,b \in M$ s.t. $\frac{a+b}{2} \in M$
of size $|M| \geq \frac{U}{2^{\sqrt{\log U}}} = U^{1-o(1)}$

so we construct this big Avg free set, do some shift to it, and intersect it with $A_1 U_1 C$
so there no internal AVG

Target 3-sum: Given S and target t ... $a+b+c=t$?
Parity 3-sum: given $S \subseteq [0, +U]$ ... $a+b=c$?